

Competition versus biology: Fertility dynamics in polygynous households in Mali

Leïla Njee Bugha*

April 14, 2026

Abstract

This paper examines whether rivalry between cowives, and wives' economic reliance on husbands influence fertility patterns in polygynous households in Mali. I test the claim from Rossi (2018) in Senegal that polygamy can be a driver of fertility. Despite the similarities between Mali and Senegal, competition between cowives after the entry of a new wife does not seem to predict fertility behavior of first wives in polygynous households in Mali. In this context, biological constraints and social constraints such as organization of marital life in polygamous households and limited female agency tend to limit fertility in polygamous households in Mali.

1 Introduction

Mali and its neighboring countries (Nigeria, Chad, Niger and Senegal) have some of the highest fertility levels in the world. World Bank data from 2018 estimates that the average

*I would like to thank Ethan Ligon for his advising, feedback and guidance throughout this project. I am thankful to Juliette Le Baron and Kalle Kappner for their support and careful proofreading, and to Abdoulaye Cissé as well as my reviewers Marco Gonzalez-Navarro, Jacob Lefler, Jeremy Magruder and Noor Sethi, for their helpful comments and suggestions.

Malian woman will have 5.9 children. Factors as varied as the low rates of contraception use, the influence of Islam, limited human capital investments for girls and women or polygyny have been mentioned to explain this phenomenon. Polygyny, a typical family structure across sub-Saharan Africa, where 11% of the population in sub-Saharan lives in households that include more than one spouse, is especially common in the region: in Mali, 34% of individuals live in a polygynous household (*Religion and Living Arrangements Around the World* 2019).

Polygyny is often assumed to drive total fertility up but despite the prevalence of these family arrangements in some regions of sub-Saharan Africa, not much is actually known about how intrahousehold dynamics play out in these households and how they might affect fertility outcomes. A recent paper by Rossi (2018) explores dynamics between husband and wives, and between co-wives within these households and shows that these relations are central in determining socioeconomic outcomes for these families. This in return generates many implications of such family arrangements for the intergenerational transmission of poverty. In the same paper, Rossi points to rivalry between cowives as a driver of fertility in polygynous households. Rossi argues that cowives compete on the number of children as it is the main way that they can extract and ensure resources for themselves from their husband due to Islamic inheritance norms and limited economic independence for women. Despite codified family codes and laws, Islamic inheritance practices favor sons, with daughters receiving only half the share of what sons are given. Wives are further disadvantaged as despite being entitled to a share equal to that of their children, they often receive one eighth of the inheritance due to the influence of Islamic and customary patrilinear laws and are especially likely to be excluded from any inheritance composed of illiquid assets, such as land or house (Lambert, Ravallion, and Walle 2011; Lambert, Walle, and Villar 2017).

Using three waves of Demographic and Health Surveys (DHS), I try to identify determinants of fertility in Mali, another West African, majority-Muslim country with a high

prevalence of polygamous households (31.2% of bigamous marriages according to 2018 DHS) and high levels of female fertility (6.3 children per woman, according to 2018 DHS). My study asks whether co-wife rivalry can generate similar fertility competition between cowives in Mali. Following Rossi (2018), I test whether fertility competition can also be observed in Mali by looking at the total number of children per wife and birth spacing before and after the entry of the second wife. I go further by trying to identify which characteristics of the wives determine fertility behavior of the first wife.

In contrast to Rossi’s results in Senegal, I find evidence that in polygynous households, women have fewer children than in monogamous households overall and there is little to no evidence of cowife rivalry pushing fertility up. Most of that difference comes from a lower fertility (as in the number of children per wife) from lower-ranked polygynous spouses, and that difference disappears once controlling for the fact that the timing of the entry of the second wife, along with the age of the husband and natural constraints on women’s fertility, will bound the number of children that can come out of second unions.

Furthermore, in the Malian setting, despite differential fertility in polygynous versus monogamous households, the competition channel does not seem to be a strong predictor of fertility outcomes for wives in polygynous households. In my results, first wives do not strongly respond to the entry of a second wife by reducing their birth spacing. Juxtaposing my findings with other findings on the distributions of ages at marriage across senior and junior wives, and between spouses, my results imply a limited scope for an interaction between the first wife’s and the cowife’s fertility behaviors, but point to social and biological constraints as major predictors of fertility behavior in Mali for polygynous and monogamous wives. Although polygyny could mechanically reduce each wife’s time spent with her husband, and thus limit reproduction, using data on sexual activity, I do not find that to be the case here. The distribution of ages shows that biological constraints (after 45 years old for women, 60 years old for men) reduce fertility for all women in general, but especially so for second wives. For polygynous women, I moreover hypothesize that restrictive gender norms

(as well as endogenous sorting and selection on the husband’s type and by the husband) might affect fertility more than rivalry or fertility competition with co-wives.

My study is directly related to studies looking at how women’s economic dependence on their husbands can directly affect their fertility decisions. Lambert and Rossi (2016) shows for example that pressure to have children and especially sons is driven by women’s need for widowhood insurance as sons can act as old-age support mechanisms, especially for women with no independent housing. Similarly, Mbaye and Wagner (2017) highlights fertility pressure for women whose family received higher bride price payments in Senegal. This strand of the literature explores how cultural factors (such as inheritance mechanisms and bride prices) can constrain women’s choices and fertility decisions, even in monogamous households. Adding to this framework, polygyny, which is often characterized by problems of limited cooperation and coordination within households (Heath, Hidrobo, and Roy 2020; Hidrobo, Hoel, and Wilson 2020), can create settings where wives end up competing or cooperating for limited resources (Reynoso 2019; Mammen 2019; Lees et al. 2021).

Previous work points to selection into polygyny and shows that differences in wives’ characteristics depending on their rank can govern the allocation of resources within the household (Akresh, Chen, and Moore 2016; Munro et al. 2010) and the investments in children (Golan and Isopi 2018). Other papers consider cooperation or complementarities between the wives in polygynous households, due to active sorting on the marriage market by the husbands and the first wives (Reynoso 2019, Akresh, Chen, and Moore 2016). This paper directly speaks to how selection or endogeneity into polygamy can affect outcomes at the household level.

2 Model and estimation strategy

2.1 Theoretical background

Previous research (Udry 1996, Duflo and Udry 2004) in West Africa has shown that resource allocation within the households is inefficient; there is also evidence of information asymmetries between spouses (Hoel 2015), with consequences on fertility behavior (Ashraf et al. 2020). In polygynous households, these inefficiencies can be further exacerbated by coordination and cooperation failures (Hidrobo, Hoel, and Wilson 2020), and power imbalances (Heath, Hidrobo, and Roy 2020; Reynoso 2019). Crucially for this paper and Rossi’s model, in polygynous households, the number of children directly determines the share of resources of each wife receives from their husband. Rossi’s model is thus meant to capture how decisions about fertility are made in such households, using a non-cooperative framework. The model is based on three mechanisms through which polygamy can affect fertility:

- Strategic complementarity: anthropological studies have long established cowife rivalry in polygamous households and conclude that these conflicts are usually caused by competing reproductive interests. Children are perceived as a sign of a husband’s preferences and interest for a specific wife but they also provide an insurance mechanism to widows and an access to the husband’s resources. Polygyny thus would lead to increased fertility due to competition between cowives.
- Strategic substitutability: occurs when fertility behavior is guided by a husband’s preferences or households’ budget constraint. Since for men, the marginal utility of children is decreasing, and men focus on the overall number of children, rather than the distribution of children between wives.
- Non-strategic reduction: refers to biological limitations on fertility, due to limitations on the time the husband can spend with each wife, as well as the natural bound in fertility caused by the higher age at marriage of husbands and wives for second and

later wives. This creates a situation where social norms (rotation rules for time spent with the husband) and biological constraints limit fertility more in polygamous than in monogamous households.

Rossi proposes a static model, in which, during her reproductive period, the woman chooses a birth rate that minimizes the distance between her ideal fertility rate, her husband's ideal fertility preferences and a natural birth rate, with the idea that deviating from this natural birth rate incurs a cost¹. The wife's objective function is formalized below, and she chooses the birth rate λ to maximize the following objective function:

$$u(n) = -(n - n_w^{id})^2 - \theta^h(n - n_h^{id})^2 - \theta^n((\lambda - \lambda_m^{nat})T)^2 \quad \text{s.t.} \quad n = \lambda T$$

$$\Rightarrow n^{NS} = \frac{n_w^{id} + \theta^h n_h^{id} + \theta^n n^{nat}}{(1 + \theta^h + \theta^n)}$$

where n_j^{id} represents the fertility preferences of the husband and wife, with $j = \{h, w\}$ and n^{nat} represents the number of children under a natural fertility regime. T represents the length of the reproductive period and λ_{nat} the natural birth rate. θ^h and θ^n capture the intensity of marital and natural constraints, i.e. penalties from deviating from the husband's ideal fertility and from deviating from the natural birth rate (such as the health or psychological cost of pursuing contraceptives) respectively.

In polygynous households, the wife's objective depends both on her own preferences, and also on the number of children born to her co-wife. For husbands, their preferences are expressed over their total number of children across all their wives, and children are perfect substitutes. Due to time and biology constraints, the natural birth rate is assumed to be lower in polygamous households than in monogamous households.

1. The birth intervals stay within a feasible range; when women reduce their birth spacing, they incur a health cost and a psychological cost from transgressing social norms. Similarly, when women increase their birth spacing beyond the range, they would incur a time or financial or psychological cost as in Ashraf, Field, and Lee (2014) from using modern contraception methods.

In the case of polygyny, the model is represented by a Nash problem, where the first and second wife simultaneously choose their birth rate to maximize their objectives. The first wife experiences two stages: a monogamous stage with the husband where her fertility rate is determined as in the monogamous case above. After this first stage, depending on the husband's type, the couple remains monogamous or the husband takes another wife, triggering the polygamous stage. The first wife then has perfect information on her own reproductive period, her own and her husband's fertility preferences. She forms expectations about her husband's type, and her cowife's fertility preferences, length of reproductive period as well as the timing of entry of the second wife. These are determined prior to the second marriage and the uncertainty is fully resolved after the second marriage takes place². Both wives choose their birth rate based on the following objective functions. In the polygamous stage, the second wife chooses her birth rate λ_2 and the first wife chooses λ_1 (with λ_0 being her birth rate chosen in the monogamous stage).

$$u_i(n_i, n_{-i}) = -(n_i - \gamma_i n_{-i} - n_{w,i}^{id})^2 - \theta^h (n_i + n_{-i} - n_h^{id})^2 - \theta^n (n_i - n_{-i}^{nat})^2$$

where $n_1^{nat} = \lambda_m^{nat} S + \lambda_p^{nat} (T_1 - S)$ and $n_2^{nat} = \lambda_p^{nat} T_2$. n_{-i} represents the fertility preferences of the other wife, and γ_i is a woman-specific parameter for co-wife rivalry intensity. λ_k^{nat} represents the natural birth rate in each stage $k = \{m, p\}$, monogamous or polygynous. S represents the length of the monogamous stage, and T_1, T_2 represent the reproductive periods for the first and second wives respectively. Solving the equilibrium:

$$n_i^* = (n_i^{NS} + n_{-i}^{NS} B_i) \times \frac{1}{1 - B_1 B_2}$$

n^{NS} represents the optimal fertility in the absence of strategic interactions, and $B_i =$

2. These characteristics as well as the incidence of polygamy are considered to be uncorrelated from fertility behavior in the monogamous stage.

$\frac{\gamma_i - \theta_i^h}{1 + \theta_i^h + \theta_i^n}$ the strategic response of wife i . The strategic response is thus increasing in γ_i (co-wife rivalry intensity) and decreasing in θ_i^h . Depending on the sign of B_i , first wives' fertility will be increasing in the fertility of their co-wives when $B_i > 0$, i.e. co-wife rivalry is intense enough to overcome the weight given to the husband's preferences in a wife's determination of her fertility. Furthermore, Rossi shows that the difference in first wives' optimal birth rates before and after the second marriage can be decomposed into a negative natural effect, a strategic effect that depends on the sign of B_i , and a negative substitution effect.

From this model, Rossi predicts that if and only if the strategic response is positive: 1) Second wives' fertility will increase with the duration of the monogamous period and time left before the end of the reproductive period of the first wife; and 2) The final number of children for both wives will increase with each wife and her co-wife's reproductive period. In that case, the competition effect is large enough to overpower the negative substitution and natural effects. These predictions that Rossi tests in her paper use longitudinal data and exits of wives to mitigate concerns over the endogeneity of the second marriage.

2.2 Discussion of the model

Rossi's model assumes that all children are substitutable across wives from the husband's point of view. This opposes evidence from many studies showing that preferences or status differences between different cowives and especially between the first wife and the other wives determines differential investment in health and education for children of each wife (Golan and Isopi 2018), different allocation and distribution of resources (using lab games in Senegal: Hidrobo, Hoel, and Wilson 2020) or differential exposure to intimate partner violence in Mali (Heath, Hidrobo, and Roy 2020). These differences imply that either husband preferences or social norms (in particular, the preeminence of the first wife) end up shaping fertility outcomes more than the model would predict. Furthermore, children both as an insurance source or as means to extract resources from the husband might also have a decreasing utility for women: even under a competition framework, a wife might find lower utility to her fifth

son, compared to her first or second son.

A second point of contention concerns non-strategic reduction: although husbands must share their time between an increasing number of wives, this does not necessarily need to decrease reproductive time. Given gender dynamics within the household, and high prevalence of sexual violence in Mali, husbands who wish to have more children can impose it to their wives. Additionally, there is an obligation for the husband under Islamic law to equally share his time between his wives. Based on data on sexual activity in the DHS, I see indeed little evidence that wives in Mali have unequal access to their husband (figure 3 in Appendix). The distribution of the number of days since last intercourse is similar across different polygynous wives (first, second and third wives have the same mean and the higher mean for fourth wives is likely due to small sample imprecision), as well as compared to monogamous wives in figure . There is however a notable difference in the probability and frequency of intercourse in the past 7 days depending on marriage type, which is 9 percentage points higher for polygynous wives (83% versus 74.4% in monogamous wives)³. If social and biological constraints are to be believed, intercourse frequency should be greater among monogamous couples than polygynous couples, which is not at all what we observe here. Two potential explanations are: either there is a social desirability bias that increases reports among polygynous couples due to co-wife rivalry, or it could be due to endogenous selection into polygamy, with polygamous men or wives more willing to engage in marital duties. I cannot test either hypothesis with my current data. Moreover, the high averages observed in the dataset do not allow me to conclude whether husbands have differential preferences to spending time with their wives in Mali, and whether these preferences are determined by spousal rank. There is little variation overall in time since last intercourse across spousal rank, and the median number of days being 7.

3. This is not due to differential age distribution between monogamous and polygamous households. Looking at wives below 45 years old, the difference remains at 9 percentage but increases to 12 points for wives above 45 years old.

2.3 Estimation

Following Rossi, I estimate two types of regressions, one looking at the total number of children per wife for women above 45 years old, and another one using dates of birth of the children and dates of marriage to study the change in birth spacing in the polygamous stage, i.e. after the entry of the second wife into the household. Due to data limitations, I am not fully able to control for the endogeneity of the second marriage⁴. My first estimating equation is given by:

$$n_{ijk} = \beta_1 T_{ijk} + \beta_2 \text{senior}_{ijk} + \beta_3 \text{junior}_{ijk} + \delta_j + \nu_k + \epsilon_i$$

where n_{ijk} is the number of (living) children per wife (self-reported), δ and ν are fixed for the cluster (census enumeration zone) and the survey wave respectively, and T_{ijk} is an indicator for being part of a polygamous marriage. In practice, rank is captured by two dummies, *senior* being a “senior” wife (first wife in a polygamous marriage) and *junior* for a “junior” wife (second wife in a polygamous marriage).

The second estimating equation is similar, except the dependent variable D_{lijk} is the birth interval (number of months between two births) and the dataset is organized at the mother-birth level (l). T_{ijk} becomes 1 for births that occurred during the polygamous stage, i.e. after the entry of the second wife in the household. T_2 represents the cowife’s reproductive period, which is defined as the number of fertile years (i.e. the minimum of the difference between 45 and the age at marriage for each wife, and the difference between 60 and the age at marriage for husbands) following Rossi (2018); X_{ijk} are woman-specific controls, including age and squared age at birth, categorical age variables for the husband and wife’s current age and λ_l are birth rank dummies. I perform a OLS and a Cox hazards regression for this

4. Rossi is able to use exits of wives from households to check the robustness of her findings.

second equation:

$$D_{lijk} = \beta_1 T_{lijk} + \beta_2 T_{lijk} * T_2 + X_{ijk} + \delta_j + \nu_k + \lambda_l + \epsilon_i$$

All the specifications reproduce the analysis and specifications chosen by Rossi, within the constraints of my data.

3 Data

I use the Demographic and Health Surveys datasets collected in Mali in 2006, 2012-13 and 2018. They contain stratified samples of households in which both men and women are interviewed. The DHS contains detailed data on fertility preferences of men and women, asking respondents about their preferred number of children by sex, along with full birth histories for women aged 15-49 who are interviewed in the Individual questionnaires containing complete birth histories, including dead children and children who moved out. The DHS interviews different women in the household but I focus on women either married or in a union, for which we have information about their current partner as well as their children together using the Couples questionnaire. I show in table 9 the summary statistics for the overall universe of women in the DHS, and for the Couples questionnaire only.

3.1 Sample selection issues

Although the overall DHS sample is representative at the national level⁵, the Couples sample does not fully mirror the universe of couples in the DHS. Most of the differences stem from the fact that the Couples sample only contains currently married or partnered respondents. (I present differences between the main sample and the full sample in Appendix in table 9).

5. I do not include the sampling weights in my calculations so the results cannot be considered representative at the national level. I show however in section A that my results are robust to the inclusion of the survey weights and remain qualitatively similar even when representative at the population level.

I focus thereafter exclusively on the Couples sample as the main sample I will use for my analysis.

Although the Couples questionnaire is meant to interview all wives in a household, there might be non random selection in who is successfully interviewed. This potentially affects my analysis as not all wives are observed, and the probability that all women of a particular man are observed decreases with the number of wives, which particularly affects polygynous households. In particular, we observe in table 1 that monogamous households tend to be over-represented in the realized interviews (96% of them are present the day of the interview), whereas 12% of first wives in polygamous households are not successfully interviewed, and 14% of second wives. For bigamous men, in only 79.4% of cases are both wives actually interviewed. There is also an additional concern that DHS does not survey non-coresiding spouses. Their sampling only includes women aged 15-49 who have stayed in the household the night before the survey or usually live in the household. This induces non random selection as non-coresiding spouses in polygynous households, who likely live close to their husband and cowives and whose fertility behavior might be affected by their cowives or vice versa, are not included in this sample. I assume here that wives who are not part of the household of their husband can be considered non-resident wives. Monogamous wives who live outside the household of their husband are not included in the Couples sample at all, but for men with only two spouses, in only 1.4% of cases is the first wife not residing with her husband and thus not interviewed; and in 18.6% of cases, the second wife is not part of the household and not interviewed. If the characteristics of the coupling and fertility are similar among resident and non-resident cowives, this should not affect the validity of my results⁶. Furthermore, the age range for eligibility in the DHS excludes women above 50 but

6. This assumption might not hold in the urban context: non-resident wives might be more educated, more independent, and less economically dependent on their husband. This would suggest that they have fewer incentives to increase their fertility as they don't need to compete for resources and are less exposed to cowife rivalry and thus would further drive down my estimates. On the other hand, Rossi finds that more educated wives, urban wives and wives whose husband has experienced a (self-reported) positive income shock exhibit stronger strategic response in response to the entry of a new wife. In the rural context, it is

they are the ones for which we would be more likely to observe their completed reproductive histories.

Table 1: Sample composition for CR questionnaire

<i>Variables</i>	N	Mean	Std. Dev.
Polygamous	8633	0.345	0.475
Number of wives	8633	1.386	0.59
Number of wives present for the interview	8633	1.302	0.531
% of monogamous wives present for interview	5652	0.962	0.191
% of first wives present for interview	1653	0.886	0.318
% of second wives present for interview	1382	0.863	0.344
All wives present for interview	8633	0.925	0.264

3.2 Summary statistics

The final sample sizes (number of complete unions interviewed by family structure) are thus 7548 couples for all three waves, corresponding to 7548 men and 8658 women.

Since the DHS is a wide-ranging survey, it also contains information about the household's economic outcomes (wealth index), individual characteristics (education), and women's economic outcomes that can be used as controls. Measurement error around measuring children's ages under 5 (which are directly surveyed) and misreporting of months of birth in DHS data that we assume to be random would bias my estimates but not threaten the validity of my analysis as I use these directly in order to compute the birth spacing for each specific woman.

Below I present summary statistics on marriage and fertility behavior among monogamous and polygynous households. Table 2 reveals that wives in monogamous and polygynous households appear to be systematically different in terms of age at the time of the interview,

not clear how different characteristics of these non-resident wives might be, as there is more homogeneity in general and non-resident wives might have been acquired differently (through levirate for example).

educational background, household resources and religion. Wives in polygamous households appear to have one more child on average than wives in monogamous households, which translates to 0.6 additional living child. Their fertility preferences are also significantly different at approximately 6 children per wife for monogamous wives, and 6.5 children per wife for polygynous wives. There is a sizable gap between monogamous and polygynous households of almost 4 children in fertility preferences of the husband, whereas the difference (still significant) is of 0.4 children for women. However, ages at marriage seem to be similar across monogamous and polygamous households at 17.1 and 16.9 years old, and at 26 and 29 years old for women and men respectively. One should note that the average age at marriage for polygamous households is likely overstated as it includes later wives who get married at later ages.

Selection in the type of men who engage in polygamy might be occurring here. Notably, polygynous households in this sample also seem to be more likely to belong to the two bottom quintiles of the wealth index distribution^{7 8}.

The average woman interviewed is aged between 28 and 31 years old, and will have more children before she is no longer fertile. A quick tabulation (see figure 4 in appendix) shows that the proportion of women aged 45 years old and above that had a child in the past 12 months preceding the survey is 3.2%, whereas the proportion of women aged 31-32 who had a child in the year preceding the interview is between 31.9 and 38.2%. When limiting the sample to women aged 45 and above, who have completed their fertility, in table 3, the differences in terms of fertility between monogamous and polygamous spouses are smaller⁹. There is a higher proportion of Muslim women in the polygynous group, and they seem

7. From the literature it is not clear where in the income distribution polygynous households should be found. Polygamy may be more prevalent in richer families, due to the economic cost incurred by the husband for multiple bride prices. Reversely, some studies find that polygamous households are poorer on average.

8. The wealth index is computed for the whole household, and only their quintile in the distribution, not the raw values for the calculated wealth index, is available in the DHS.

9. Women in polygynous households appear to be more likely to be infertile, but this is likely endogenous as infertility will lead the husband to marry another wife in order to have children.

Table 2: Marital characteristics and fertility preferences by polygamy status

	Monogamous			Polygamous			Test
	N	Mean	SD	N	Mean	SD	
Bottom quintiles wealth index	5652	0.375	0.484	3006	0.458	0.498	F= 56.698***
Muslim	5633	0.926	0.262	2999	0.939	0.239	F= 4.99**
Current age - W	5652	28.717	7.909	3006	31.759	8.035	F= 287.29***
Years of education	5649	1.638	3.397	3002	0.741	2.198	F= 171.266***
Received any education	5652	0.236	0.425	3006	0.134	0.34	F= 129.977***
Current age - H	5652	38.222	9.375	3006	43.937	8.293	F= 788.786***
Age at first birth	5250	18.577	3.757	2835	18.642	3.997	F= 0.527
Age at marriage	5652	17.127	3.83	3006	16.925	3.893	F= 5.362**
Husband's age at marriage	5652	26.627	6.116	3006	29.075	7.48	F= 268.252***
Number of children (ever born)	5652	3.708	2.575	3006	4.6	2.811	F= 220.933***
Number of children (living)	5652	3.176	2.134	3006	3.732	2.269	F= 127.412***
Ideal number of children - H	4690	7.845	4.784	2315	11.749	6.205	F= 842.621***
Ideal number of children - W	5021	6.094	2.494	2643	6.482	2.471	F= 42.132***
Ideal number of boys - W	5014	2.924	1.673	2641	3.119	1.681	F= 23.31***
Ideal numbers of girls - W	5014	2.529	1.481	2641	2.707	1.485	F= 24.938***
Ideal number is zero - W	5021	0.018	0.133	2643	0.017	0.131	F= 0.051

Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Bottom quintiles refer to the wealth index, which is measured at the household level adjusting for differences between rural and urban households. "H" and "W" indicate that these outcomes are referring to husbands, and wives respectively. "Ideal number of children - H" is the ideal number of children as expressed by the husband.

to have 0.5 child more on average than monogamous wives looking at their entire birth histories. Differences that might relate to the endogenous selection into polygamy by the husband or the wife or be the result of polygyny (such as religion, education, and other individual characteristics) still remain. The number of children per wife also seems to be lower under polygyny, for wives who have completed their reproductive histories.

4 Results

4.1 Fertility behavior and marriage type

The summary statistics document a marriage age gap between husbands and wives, in both types of households. I show in the table 4 as well as in figure 1 that for first and second wives, the age at marriage seems to have overlapping distributions with a slightly lower mean

Table 3: Marital characteristics and fertility preferences by polygamy status - for women 45 years old and above

	Monogamous			Polygamous			
	N	Mean	SD	N	Mean	SD	
Bottom quintiles wealth index	251	0.47	0.5	243	0.473	0.5	F= 0.005
Muslim	251	0.892	0.31	243	0.959	0.199	F= 7.955***
Current age - W	251	46.438	1.388	243	46.272	1.327	F= 1.858
Years of education	251	1.239	3.105	243	0.761	2.245	F= 3.819*
Received any education	251	0.171	0.378	243	0.136	0.343	F= 1.194
Current age - H	251	53.61	5.49	243	52.807	3.79	F= 3.558*
Age at first birth	247	20.879	5.415	234	21.107	5.623	F= 0.206
Age at marriage - W	251	19.084	5.388	243	18.407	4.677	F= 2.214
Age at marriage - H	251	25.801	5.928	243	24.848	5.613	F= 3.362*
Has no children	251	0.02	0.14	243	0.053	0.225	F= 3.981**
Number of children (ever born)	251	6.972	2.959	243	6.593	3.169	F= 1.894
Number of children (living)	251	5.614	2.439	243	5.189	2.538	F= 3.588*
Ideal number of children - H	194	8.655	5.268	187	12.422	7.062	F= 35.015***
Ideal number of children - W	204	6.961	3.093	198	6.788	2.693	F= 0.356
Ideal number of boys - W	203	3.281	1.933	198	3.146	1.831	F= 0.51
Ideal numbers of girls - W	203	2.808	1.703	198	2.763	1.704	F= 0.071
Ideal number is zero - W	204	0.034	0.182	198	0.02	0.141	F= 0.749

Note: “Bottom quintiles” refers to a dummy for belonging to the bottom 2 quintiles of the household wealth index, which is computed at the household level and adjusting for wealth differences across rural and urban locales. “H” and “W” indicate that these outcomes are referring to husbands, and wives respectively. “Ideal number of children - H” is the ideal number of children as expressed by the husband.

age at marriage for first wives.

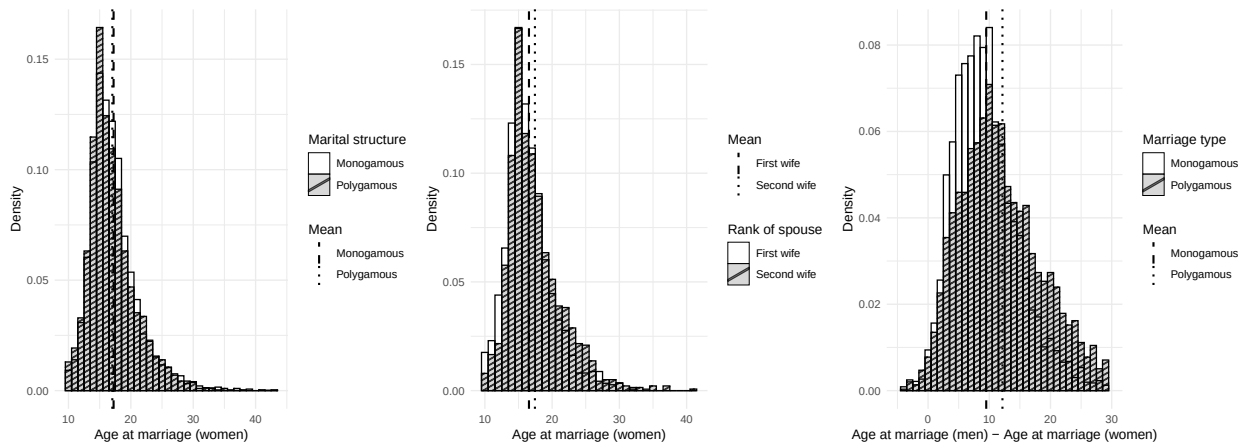


Figure 1: Age at marriage, by marriage type, by rank and between men and women

Table 4 shows that the median reproductive period is 29 years for the first wife and 28

years for the second wife. Given that the second wife comes in approximately 7 years after the first marriage, this suggests that the age difference between the two wives might not be enough to generate the fertility response observed in Rossi (2018). Since the two wives are close in age, the first wife might choose to not engage in the competition at all for three potential reasons: 1) She might not have had enough children during the monogamous stage to later keep up with a younger woman’s fertility; 2) The husband’s preferences for his latest wife after the marriage might further reduce her chances of conception and 3) their age difference is still large enough for the second wife to have more children when the first wife is no longer fertile.

The overlapping distributions of age at marriage for wives of different ranks to the same husband indicate that the number of fertile years is also going to be similar across wives within marriages. Using figure 4, it seems that 45 and 60 appear to be natural cutting points for women and men respectively.

Table 4: Timing of unions by spousal rank (polygynous spouses only)

	25th	Med.	75th	25th	Med.	75th
	First wife			Second wife		
Current age - wife	29	35	40	23	28	34
Current age - husband	38	44	50	38	45	51
Wife’s age at marriage	14	16	18	15	16	19
Wife’s age at second marriage	19	23	27	-	-	-
Husband’s age at marriage	22	25	29	27	32	37
Husband’s age at second marriage	27	32	37	27	32	37
Age at first birth	16	18	20	16	18	21
Number of children (living)	3	4	6	1	3	4
Reproductive period (years)	27	29	31	22	26	29

Thus, in order to show here that ages of the wives will tend to affect their fertility, and their fertility response, I regress the number of living children per wife on the marriage type, and on the rank and reproductive period of the wives, focusing on a sample of married women aged 45 years old and above. Women in polygynous households seem to have 0.4

fewer child (significant at the 10% level), and most of that effect comes from junior wives. Indeed when controlling for the reproductive period, the magnitude of the effect from the junior wife having fewer children shrinks. This confirms the importance of the length of the reproductive period as a biological driver of fertility. These results also allow to confirm that fertility is also partly determined by marriage type, as the number of children decreases with the rank of the wife. This remains true even when controlling for the overselection of infertile women in polygynous households in columns 3 and 4, selecting only women who have at least one child. These results are robust to the inclusion of dummies for quintiles of the household wealth index and for being a Muslim household, as well as to increasing the reproductive period for men to 65 and 70 years old, to using all children ever born as a dependent variable (see Appendix).

Table 5: Fertility patterns for women with complete birth histories

Dependent Variable:	Number of living children			
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Polygamous	-0.4052* (0.2249)			
Senior wife		-0.0729 (0.2452)	-0.0273 (0.2313)	-0.0965 (0.2261)
Junior wife		-0.7996** (0.3574)	-0.6802** (0.3326)	-0.6045* (0.3392)
Reproductive period				0.0954*** (0.0198)
Observations	494	494	476	476
R ²	0.00971	0.01246	0.0109	0.04903
Within R ²	0.00657	0.00933	0.00797	0.04622
<i>Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Fixed effects for the different survey rounds are included. Columns 3 and 4 only include women who have at least one child, discarding infertile unions. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>				

Since the magnitude of the coefficients is different by a factor of 10 between junior and senior wives, this suggests that some of the effect described by Rossi is present, with differential fertility behaviors depending on the entry and characteristics of the second wife. However, the small and negative coefficient on senior wives could imply that there is limited

selection into polygyny by women at play, since the senior wives have as many children as monogamous wives. When combining the different waves of data from DHS¹⁰, I find that results are quite different from one wave to the next. Data from the 2006 sample reproduces most closely the results from the Senegal study, which also uses data collected in 2006-2007. There is a strongly significant difference in fertility between polygynous and monogamous households: women in polygynous households who were 45 years old and above seem to have one fewer child than women in monogamous households. Controlling for the rank of the spouse, the effect remains significant and consistent, although only marginally significant when looking at wives who have at least one child. The couple's reproductive period remains significant but does not change the magnitude of the effects from the negative coefficients on both the senior and the junior wife.

In the 2012-2013 results (columns 3 and 4 of table 6), the coefficient on the polygamous dummy is statistically insignificant but positive, suggesting that women in polygamous households tend to have more children per wife than in monogamous households. The coefficient on *senior wife* across all four specifications confirms this trend; the coefficient on *junior wife* is negative, showing a potential fertility gap between senior and junior wives. The coefficient on the reproductive period is strongly significant, suggesting mostly a role for age differentials in determining fertility levels, rather than rank distributions within the household. The 2018 sample (columns 5 and 6 of table 6) shows patterns similar to the 2006 sample (with polygamous households having fewer children on average) but none of the coefficients are statistically significant and their magnitude is much smaller.

10. I added the 2006 sample at a later date to increase sample size of women with completed birth histories.

Table 6: Fertility for women who completed their birth histories (aged 45 years old and older), by DHS wave

Dependent Variable:	Number of living children					
	2006 sample		2012-2013 sample		2018 sample	
Polygamous	-0.9895*** (0.3678)		0.1979 (0.3724)		-0.4551 (0.4304)	
Senior wife		-0.6446* (0.3668)		0.5796 (0.3934)		-0.2309 (0.4181)
Junior wife		-1.062* (0.6102)		-0.2088 (0.5313)		-0.5192 (0.6060)
Reproductive period		0.0894*** (0.0339)		0.1049*** (0.0311)		0.0777** (0.0377)
Constant	5.8*** (0.2662)	3.333*** (0.9257)	5.49*** (0.2484)	2.896*** (0.7757)	5.588*** (0.2890)	3.814*** (0.9294)
<i>Fit statistics</i>						
Observations	170	164	176	173	148	139
R ²	0.04064	0.0517	0.00163	0.08895	0.00762	0.02573
Adjusted R ²	0.03493	0.03392	-0.00411	0.07277	0.00083	0.00408
<i>Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Fixed effects for the different survey rounds are included. Columns 2, 4 and 6 only include women who have at least one child, discarding infertile unions. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>						

Contrasting the results from these three different waves of data requires us to identify potential differences in the sample¹¹. For readability and simplicity, I compare the 2006 sample to the two other samples combined together in table 15 in the Appendix. Using the limited sample of married women 45 years old and older, I find substantial differences in the age at marriage for both husband and wives, and in the number of children ever born (see table 16 in Appendix for a comparison of all three rounds). There is a slight (but significant) difference in the polygamy dummy, as well as in the stated fertility preferences. The number of living children, the ideal number of children for the husband and the current age of the wife do not vary much across rounds. This test implies that the fixed effects might not adequately

11. Notably, 2012-2013 saw the emergence of an armed conflict between the government and Jihadist independence groups in the Northern regions (the conflict is still ongoing). The conflict likely disturbed the data collection, which spanned over two years and did not include the three Northern regions, Tombouctou, Gao and Kidal, and some districts of the Mopti region (for more detail, see p.5-6 of <https://dhsprogram.com/pubs/pdf/FR358/FR358.pdf>). It likely also affected marriage and fertility patterns. These differences are still observed after only analyzing the regions which were collected each time.

control for time-varying trends between the three waves of survey, since the outcomes most affected in the sample of women above 45 years old also saw a secular decline in polygamy, in the number of children, improvements in child mortality, etc. These changes might also affect bargaining positions within the household, as well as perceptions of (high versus low) fertility in some households. It is important to note that there is no change across time in the ideal number of children expressed by husbands, which hovers between 10.3 and 10.9 in the 12-year window described by the data.

4.2 First wives' birth spacing behavior

Using birth dates of different children and dates of union, I construct birth intervals for children of the first wife born before or after the entry of the second wife in the household. Column 1 from table 7 shows that the entry of a second wife does not seem to affect the birth intervals when controlling for wife's age at birth and its square, birth order dummies and categorical current ages of the husband and wife. However, there is wide heterogeneity depending on how fertile the cowife is. A fertile cowife is defined as a cowife whose reproductive period has an above-median length (26 years or longer). The first wife's birth spacing increases by 2.2 months when the new wife entering is less fertile. This provides more evidence that a wife's fertility might be partly determined by the characteristics of her cowife. The first wife slightly decreases her birth spacing when facing a more fertile cowife, by less than a month out of a mean of 31.2 months (although the coefficient is not statistically significant). Controlling for the cowife's reproductive period doesn't seem to affect the results much: an increase of the cowife's reproductive period by one year is associated with a decrease of the birth intervals by 15 days, which is not very economically meaningful. The patterns observed do not seem fully compatible with a potent competition effect, or at least not one that will be driven by classic factors of fertility: on average, the cowife's reproductive period does not change the behavior of the first wife, who reduces her fertility in response to the entry of a new wife. As a robustness check, I use a Cox hazards regression in the Appendix table 18.

These results show a slight increase in birth rates as a response to the second marriage, as a whole and for women with a less fertile co-wife. The Cox specification compares specifically what happens at the level of the birth interval during which another wife enters the household. When accounting for the cowife’s reproductive period, I find suggestive evidence that an increase in the length of the cowife’s reproductive period reduces birth rates by 5%. Table 19 allows me to rule out that any explanatory factors (education, gender of children, husband’s income), mentioned by Rossi as strengthening the strategic response can generate increases in fertility in the Malian context.

The differences in birth spacing increases depending on the fertility of the second wife also suggest that unless there is a clear signal of lower fertility from the second wife, the first wife will not compete or react to the cowife’s fertility. A clear signal here is a short reproductive period of less than 26 years, but it could also be a higher age at marriage of the second wife or a second wife that is inherited through social norms (levirate, widows from the family or village) and not the husband’s preference for a wife. There can also be heterogeneous responses depending on the fertility signals from the cowife¹² that are not adequately captured by the linear specification (column 4). The columns 2 and 3 split the sample based on the 2nd wife’s fertility period, below and above average median, showing differential (but not statistically so) responses of the first wife to the entry of the 2nd spouse, depending on the length of the 2nd spouse’s reproductive window¹³.

4.3 Other predictors of fertility

These results on birth spacing and fertility behavior, along with the evidence on the identical distributions of ages for polygynous wives in Mali point to other factors as predictive of the

12. I test whether low fertility preferences change the results and find that increasing the probability that the cowife wants less than five children is associated with a significant increase at the 10% level of birth intervals by 1.8 month (see table 21).

13. The sample is not split equally because the median is calculated on a subsample of junior wives, but the condition is assigned to the whole household, potentially inflating the "below median" category in the analysis sample.

Table 7: Comparison of birth intervals including flexible age controls

	Birth intervals (in months)			
	Whole sample	Below median	Above median	
		reproductive period		
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Polygamous stage	0.3168 (0.6626)	2.255* (1.345)	0.1774 (1.220)	15.49*** (4.073)
Polygamous stage $\times$ cowife's reproductive period				-0.5305*** (0.1456)
Control mean	31.1	30.6	31.2	31.1
<i>Fixed-effects</i>				
Wave variable	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	6,855	1,892	2,018	4,464
R ²	0.03863	0.04562	0.03431	0.04155
Within R ²	0.03303	0.03797	0.03169	0.03673

*Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Sample for this regression is all senior wives below 45 years old, with at least one child. All specifications include birth rank dummies, age and age squared at first birth for the first wife, categorical variables of the current ages of the husband and the wife in 5-year bands. Fixed effects for the different survey rounds are included. Polygamous stage refers to a dummy that takes the value 1 after the entry of the second wife in the household. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

number of children per wife. Instead of exploring the rivalry angle, I look to qualitatively identify the effect of men's preferences and other determinants on fertility. I conduct an exploratory analysis¹⁴ regressing actual fertility on ages at marriage, fertility preferences, education, number and rank of wives, frequency of intercourse in the past month.

In table 22, the strongest predictors of fertility seem to be the marriage type, as being a junior wife is associated with a decrease in 1 child per wife in fertility. The positive but small coefficient on intercourse also points in the same direction. Other significant predictors of the model include women's ages at marriage moving in the opposite direction as men's ages at marriage or women's ages at first birth. This is consistent with the fact that the variable

14. I choose an OLS estimation procedure, which is also adequate for prediction but LASSO or a factor analysis would be suited here to identify predictors of fertility for women in polygynous households

Reproductive period is determined by the wife's fertility cycle 85% of the time. The couple's reproductive period follows the number of fertile years by the woman, rather than the man, as it takes the minimum between the two, as figure 2 shows.

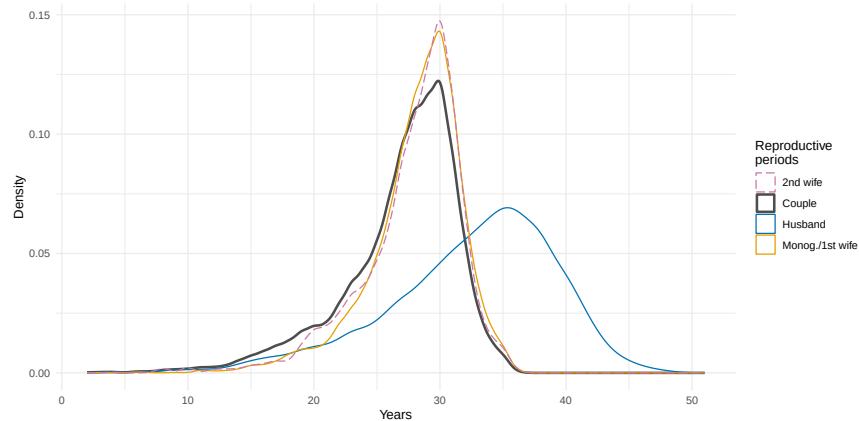


Figure 2: Reproductive periods and fertile years for men and women

The age gap between the first and second wife then determines most of the effect on fertility, as the husband's age is not the binding constraint restraining the second wife's fertility.

This, combined with the positive and significant coefficient on the woman's fertility preferences indicates that men's characteristics do not play a large role in predicting the household's fertility outcomes. Although the coefficient on men's preferences is significant, it is twice as small in magnitude as the coefficient on women's desired fertility: an increase in the ideal number of children for men is only correlated with a 2% increase in the actual number of children. The distribution of realized fertility seems to match more closely, women's fertility preferences than men's (figures 5 and 6 in Appendix). Comparing the preferences for all women and for women who have completed their reproductive histories, men's and women's fertility preferences seem to align more closely but both keep a longer right tail than realized fertility.

The coefficient on *Senior wife* being negative is consistent with table 7, which shows limited effort from the first wife to strategically respond to a cowife's expected fertility. If

anything, entry into polygamy significantly reduces the first wife’s fertility, by increasing her birth spacing. Table 22 finds no correlation between being the first wife and realized fertility, suggesting that first wives act similarly to monogamous wives, as seen in table 5.

Table 8: Predictors of fertility (dep.var: number of children per wife)

	Ever born (1)	Living (2)
<i>Variables</i>		
Senior wife	0.0597 (0.2196)	0.0380 (0.1809)
Junior wife	-1.145*** (0.2157)	-1.016*** (0.1776)
Years of education	-0.0590*** (0.0174)	-0.0433*** (0.0142)
Age at first birth	-0.1435*** (0.0121)	-0.0928*** (0.0106)
Age at marriage - W	0.0757*** (0.0133)	0.0495*** (0.0117)
Age at marriage - H	-0.0649*** (0.0063)	-0.0438*** (0.0053)
Ideal number of children - H	0.0816*** (0.0078)	0.0598*** (0.0065)
Junior wife $\times$ Ideal nb. - W	0.0008 (0.0026)	-0.0010 (0.0021)
Ideal number of children - W	0.1475*** (0.0206)	0.1098*** (0.0166)
Frequency intercourse past month	0.0106** (0.0051)	0.0123*** (0.0044)
Survey wave fixed effects	Yes	Yes
<i>Fit statistics</i>		
Observations	3,002	3,002
R ²	0.36648	0.33339
Within R ²	0.36120	0.33223

Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration area level. “H” and “W” indicate that these outcomes are referring to husbands, and wives respectively. “Ideal number of children - H” is the ideal number of children as expressed by the husband. Fixed effects for the survey waves are included, as well as controls for being Muslim and being part of the two bottom quintiles of the wealth index distribution. Missing values for ideal number of children (for both men and women), age at first birth and frequency of intercourse are imputed, and indicator variables for missingness in these variables are included in each regression.

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

5 Discussion and conclusion

Since being a lower ranked wife drives down fertility, and first wives do not seem to fully react to the “competition” of other cowives in their household, this suggests either that polygamy risk is not fully anticipated by these wives and they do not modify their birth rate in preparation for it, nor in response to it. If a new wife appears to be very fertile, the first wife does not compete; she increases her birth spacing by more than 1 year on average. Reconciling these facts with the Rossi’s model and my findings on junior wives suggests that either rivalry between cowives is not as prevalent in Mali than in Senegal; or that wives in Mali might lack the agency necessary for them to realize their desired fertility or move fertility outcomes.

Given the cultural and religious similarities between Mali and Senegal, and the fact that cowife rivalry has been well documented in Mali (Jankowiak, Sudakov, and Wilreker 2005), there is no reason to believe that there would be less rivalry in Mali but it might play out in a different area than fertility, e.g. the actual allocation of resources within the household¹⁵. Looking at potential factors that might strengthen fertility competition, I find that none of the covariates predicted to increase strategic response by Rossi are significant here.

Thus, the latter explanation seems more relevant here. Unpublished data from another project in Mali suggests that at least among the poorest rural households, wives have limited information on the features of their own marriage: 33 to 38% of wives interviewed report not knowing the amount of their dowry or how much of it has been paid at the time of the survey. Asked about their matrimonial situation, 64% of currently monogamous wives who got married civilly (18.6% of respondents in total) signed the polygamy option. Of the remaining wives married under customary law (81.4% of the total), only 17% say that they have an agreement with their husbands that he will not marry another wife, and 83% of

15. This recent paper, Lees et al. 2021, using qualitative methods in one region in Mali finds high degree of cooperation between cowives, especially in child rearing: this suggests that in some cases, children might also be substituable across cowives for women as well.

them report that they either do not know if their husband will marry another wife or that their husband will marry another wife if he wishes to. This lack of agency, which might be compounded by the lack of economic independence, might prevent women from treating their birth rate as a choice variable, as the model suggests they do. Many women in this context might not be able to enforce any sort of fertility decisions, and even less so to respond to the fertility of their cowives. Figure 2 shows that ideal fertility when combined for all wives of a specific husband follows an almost identical distribution to the ideal fertility of the husband - this would further suggest that the fertility decision might be somewhat exogenous to the woman. Extensions of this model might explore the role of the husband, and consider how husband's fertility preferences affect his choice of marriage type and of wives.

Junior wives end up having fewer children as they experience a shorter reproductive period; because children are substitutable in the eyes of the husband, they might end up substitutable as well for the wife (Lees et al. 2021); and because there are social and natural limits to their fertility as they share their husband with another. My data doesn't show conclusively that husband time sharing reduces opportunities for reproduction either. Taken together, my results imply that in the Malian context, rivalry between cowives might be insufficient to push fertility in polygynous households above levels observed in monogamous households, and counteract the downward effects of biological constraints in polygynous households.

Strategic interactions between cowives' fertility do not fully capture fertility patterns in Malian polygynous households. Using age at marriage variation and the lack of "reaction" by first wives, I make the case that social norms and biological constraints are sufficient to explain fertility of second wives in Mali. For first wives, either poverty might make having more children as a means to capture resources for women more costly than beneficial or limited agency can help explain why we see similar fertility behavior between monogamous and polygamous first wives.

Furthermore, my findings suggest that selection into polygyny should be examined at the level of the husband. Indeed, potential explanations for the lack of response from first wives involve misperceptions or erroneous anticipations by women of the risk of polygamy. Further research could explore fertility decisions within households on husbands' decisionmaking, and marital decisions and investments.

References

- Akresh, Richard, Joyce J. Chen, and Charity T. Moore. 2016. Altruism, Cooperation, and Efficiency: Agricultural Production in Polygynous Households. *Economic Development and Cultural Change* 64 (4): 661–696.
- Ashraf, Nava, Erica Field, and Jean Lee. 2014. Household bargaining and excess fertility: an experimental study in Zambia. *American Economic Review* 104, no. 7 (July): 2210–37.
- Ashraf, Nava, Erica Field, Alessandra Voena, and Roberta Ziparo. 2020. Maternal mortality risk and spousal differences in the demand for children.
- Duflo, Esther, and Christopher Udry. 2004. *Intrahousehold resource allocation in Cote d'Ivoire: social norms, separate accounts and consumption choices*. Working Paper, Working Paper Series 10498. National Bureau of Economic Research, May.
- Golan, J., and Alessia Isopi. 2018. The allocation of resources within polygamous households : evidence from Nigeria.
- Heath, Rachel, Melissa Hidrobo, and Shalini Roy. 2020. Cash transfers, polygamy, and intimate partner violence: experimental evidence from Mali. *Journal of Development Economics* 143:102410. ISSN: 0304-3878.
- Hidrobo, Melissa, Jessica Hoel, and Katie Wilson. 2020. Efficiency and status in polygynous pastoralist households. *The Journal of Development Studies* 57 (May): 1–17.

- Hoel, Jessica B. 2015. Heterogeneous households: a within-subject test of asymmetric information between spouses in kenya. *Economic Experiments in Developing Countries, Journal of Economic Behavior Organization* 118:123–135. ISSN: 0167-2681.
- Jankowiak, William, Monika Sudakov, and Benjamin C. Wilreker. 2005. Co-wife conflict and co-operation. *Ethnology* 44 (1): 81–98.
- Lambert, Sylvie, Martin Ravallion, and Dominique van de Walle. 2011. *Is it what you inherited or what you learnt? Intergenerational linkage and interpersonal inequality in Senegal*. WIDER Working Paper 2011/062. Helsinki: UNU-WIDER.
- Lambert, Sylvie, and Pauline Rossi. 2016. Sons as widowhood insurance: evidence from Senegal. *Journal of Development Economics* 120 (May): 113–127.
- Lambert, Sylvie, Dominique van de Walle, and Paola Villar. 2017. *Marital trajectories and women’s wellbeing in Senegal*. WIDER Working Paper 2017/124. Helsinki: UNU-WIDER. <https://doi.org/10.35188/UNU-WIDER/2017/350-9>.
- Lees, Shelley, Nambusi Kyegombe, Ampa Diatta, Amanda Zogrone, Shalini Roy, and Melissa Hidrobo. 2021. Intimate partner relationships and gender norms in mali: the scope of cash transfers targeted to men to reduce intimate partner violence. *Violence Against Women* 27 (3-4): 447–469.
- Mammen, Kristen. 2019. All for one or each for her own: do polygamous families share and share alike? *Journal for the Advancement of Developing Economies* 8 (1): 2161–8216.
- Mbaye, Linguère Mously, and Natascha Wagner. 2017. Bride price and fertility decisions: evidence from rural Senegal. *The Journal of Development Studies* 53 (6): 891–910.
- Munro, Alistair, Bereket Kebede, Marcela Tarazona-Gomez, and Arjan Verschoor. 2010. *The lion’s share. An experimental analysis of polygamy in Northern Nigeria*. GRIPS Discussion Papers 10-27. National Graduate Institute for Policy Studies, December.
- Religion and Living Arrangements Around the World*. 2019. Report. Pew Research Center, December.

- Reynoso, Ana. 2019. Polygamy, co-wives' complementarity and intra-household inequality. Unpublished.
- Rossi, Pauline. 2018. Strategic choices in polygamous households: theory and evidence from Senegal. *The Review of Economic Studies* Volume 86, no. 3 (May): 1332–1370.
- Udry, Christopher. 1996. Gender, agricultural production, and the theory of the household. *Journal of Political Economy* 104 (5): 1010–1046. ISSN: 00223808, 1537534X.

A Appendix

A1. Additional graphs

Figure 3: Intercourse frequency by spousal rank

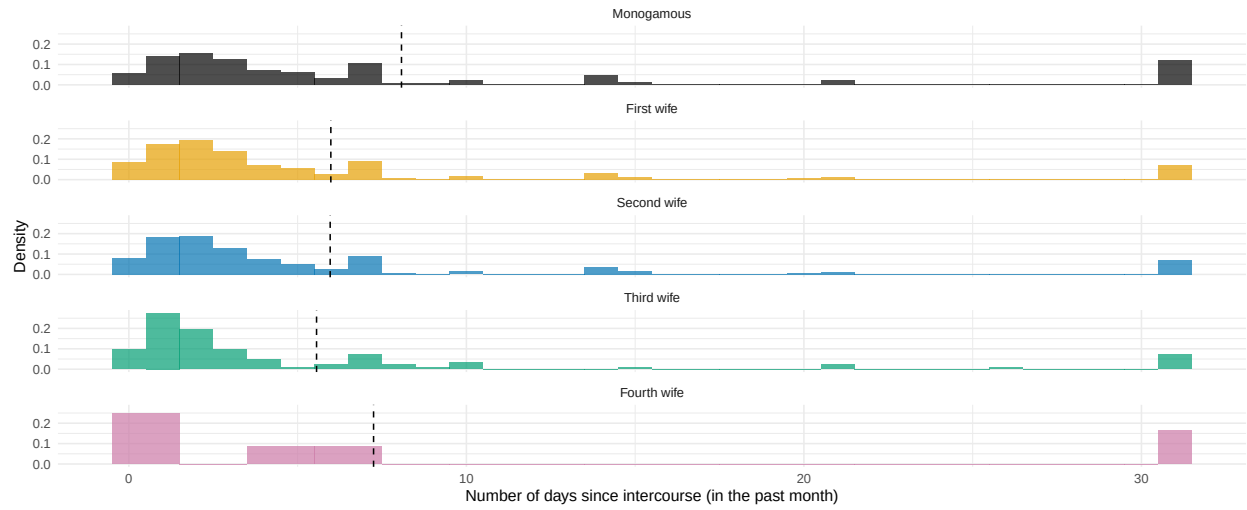


Figure 4: Likelihood of a birth in past year by age

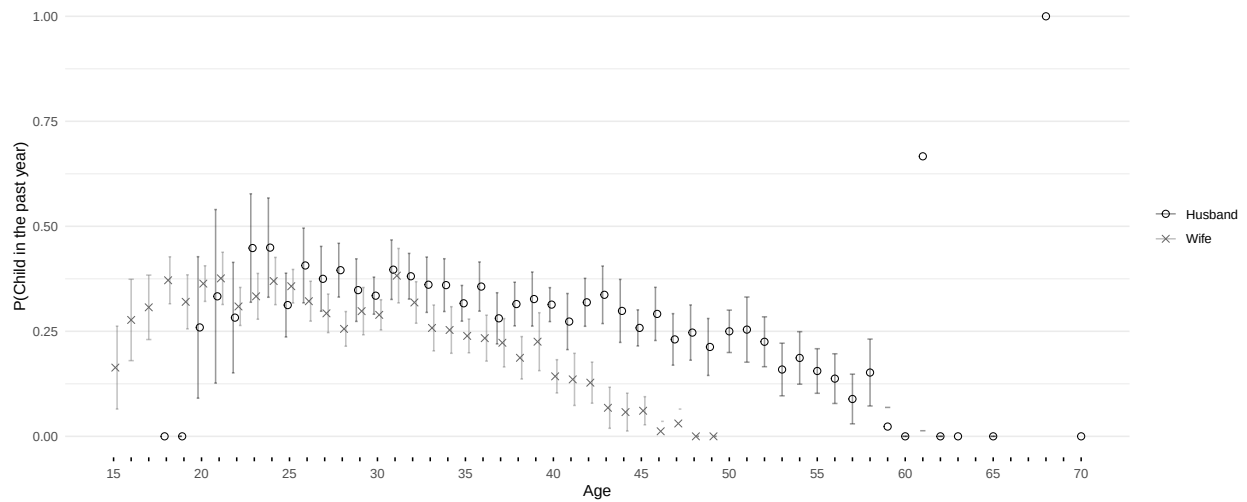


Figure 5: Fertility preferences and fertility outcomes as $\log(1 + n)$

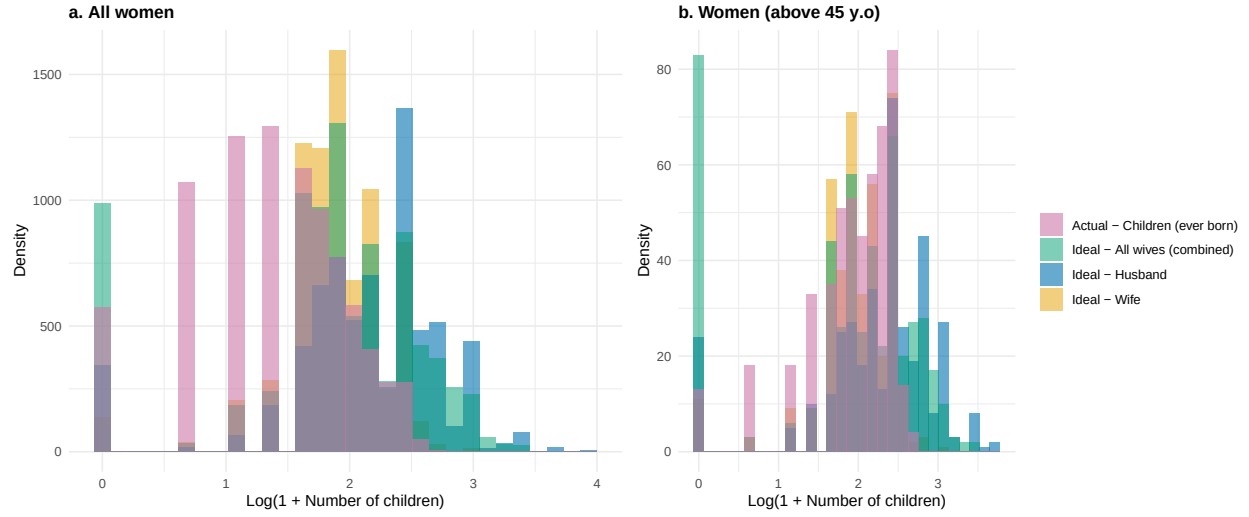
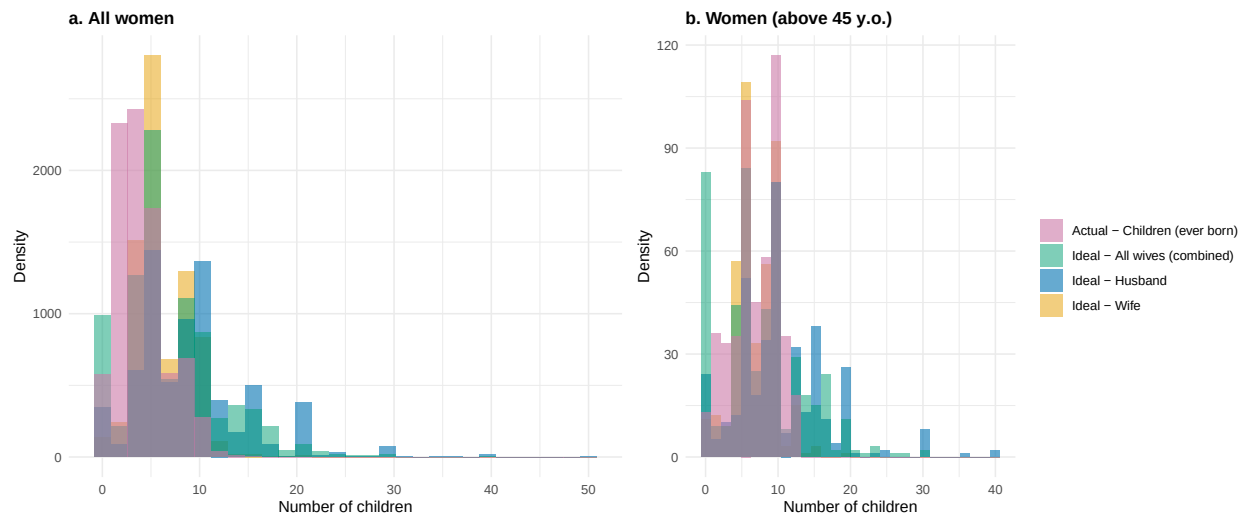


Figure 6: Fertility preferences and fertility outcomes



A2. Additional tables

Table 9: Demographic characteristics of the sample

table/header-rows=1,2						
<i>Variables</i>	Full sample			Couples sample		
	N	Mean	SD	N	Mean	SD
Currently in a union	35526	0.827	0.378	8658	1	0
Current age	35526	28.405	9.219	8658	29.773	8.083
Age at marriage	30484	17.155	3.958	8658	17.162	3.85
Husband's age at marriage	29351	28.864	8.662	8658	25.654	5.774
Polygamous wife	35526	0.295	0.456	8658	0.347	0.476
Years of education	35475	1.905	3.594	8651	1.327	3.065
Any education	35526	0.267	0.443	8658	0.201	0.4
At least one child	35526	0.78	0.415	8658	0.921	0.27
Children ever born, #	35526	3.359	2.926	8658	4.018	2.693
Children (living), #	35526	2.789	2.386	8658	3.369	2.198
Ideal number of children	31537	6.032	2.573	7664	6.228	2.493

Table 10: Demographic characteristics of the sample - in union only

table/header-rows=1			
Variable	N	Mean	Std. Dev.
Currently in a union	29393	1	0
Current age	29393	29.999	8.653
Age at marriage	29393	17.137	3.931
Husband's age at marriage	29351	28.864	8.662
Polygamous wife	29393	0.356	0.479
Years of education	29360	1.435	3.189
Any education	29393	0.212	0.408
At least one child	29393	0.897	0.304
Children ever born, #	29393	3.918	2.81
Children (living), #	29393	3.256	2.284
Ideal number of children - W	25984	6.212	2.561

Table 11: Fertility for women 45+ (completed fertilities) - dep. var: children ever born

Dependent Variable:	Number of children ever born			
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Polygamous	-0.4691*			
	(0.2736)			
Senior wife		-0.0750	-0.0317	-0.1406
		(0.2994)	(0.2839)	(0.2720)
Junior wife		-0.9164**	-0.7665*	-0.6473
		(0.4609)	(0.4346)	(0.4438)
Reproductive period				0.1503***
				(0.0243)
<i>Fixed-effects</i>				
Wave variable	Yes	Yes	Yes	Yes
Observations	494	494	476	476
R ²	0.02647	0.02878	0.03408	0.09576
Within R ²	0.00593	0.00828	0.00676	0.07019
<i>Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Standard errors are clustered at the census enumeration areas. Fixed effects for the different survey rounds are included. Columns 3 and 4 only include women who have at least one child, discarding infertile unions.</i>				

A3. Additional tables with sampling weights

Table 12: Fertility for women 45+ (completed fertilities) - controls

Dependent Variable: Model:	Number of living children			
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Polygamous	-0.4192* (0.2254)			
2nd quintile of wealth index	-0.1663 (0.3301)	-0.1709 (0.3303)	-0.0939 (0.3082)	-0.1067 (0.3043)
3rd quintile of wealth index	0.0294 (0.3500)	-0.0248 (0.3490)	-0.1680 (0.3364)	-0.1603 (0.3308)
4th quintile of wealth index	-0.1079 (0.3438)	-0.1741 (0.3440)	-0.2232 (0.3241)	-0.2919 (0.3168)
5th quintile of wealth index	-0.4616 (0.3638)	-0.4738 (0.3648)	-0.3331 (0.3389)	-0.2336 (0.3260)
Senior wife		-0.0891 (0.2459)	-0.0250 (0.2313)	-0.0850 (0.2266)
Junior wife		-0.8112** (0.3577)	-0.7028** (0.3348)	-0.6205* (0.3403)
Reproductive period				0.0955*** (0.0201)
Observations	494	494	476	476
R ²	0.01416	0.01666	0.01326	0.05098
Within R ²	0.01103	0.01354	0.01034	0.04817
<i>Statistical significance markers: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. The wealth index is measured at the household level adjusting for differences between rural and urban households, and divided into quintiles. Standard errors are clustered at the census enumeration areas. Fixed effects for the different survey rounds are included. Columns 3 and 4 only include women who have at least one child, discarding infertile unions.</i>				

Table 13: Fertility for women 45+ (completed fertilities) - controls

Dependent Variable:	Number of living children			
Model:	(1)	(2)	(3)	(4)
Polygamous	-0.3240 (0.2250)			
Islam dummy	-1.181*** (0.3612)	-1.216*** (0.3628)	-1.034*** (0.3582)	-1.08*** (0.3475)
Senior wife		-0.0019 (0.2451)	0.0343 (0.2312)	-0.0333 (0.2255)
Junior wife		-0.7161** (0.3550)	-0.6086* (0.3306)	-0.5284 (0.3365)
Reproductive period				0.0970*** (0.0195)
Observations	494	494	476	476
R ²	0.02492	0.02865	0.0251	0.06452
Within R ²	0.02183	0.02557	0.02222	0.06175

*Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Fixed effects for the different survey rounds are included. Columns 3 and 4 only include women who have at least one child, discarding infertile unions. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 14: Fertility for women 45+ (completed fertilities)

Dependent Variable:	Number of living children				
Model:	(1)	(2)	(3)	(4)	(5)
Polygamous	-0.4052* (0.2249)				
Senior wife		-0.0729 (0.2452)	-0.0273 (0.2313)	-0.1030 (0.2254)	-0.1056 (0.2251)
Junior wife		-0.7996** (0.3574)	-0.6802** (0.3326)	-0.6061* (0.3399)	-0.6098* (0.3402)
Reproductive period till 70				0.1009*** (0.0201)	
Reproductive period till 65					0.1032*** (0.0201)
Observations	494	494	476	476	476
R ²	0.00971	0.01246	0.0109	0.05404	0.05662
Within R ²	0.00657	0.00933	0.00797	0.05124	0.05383

*Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Fixed effects for the survey rounds are included. Columns 3, 4 and 5 only include women who have at least one child. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 15: Means comparison for samples of married women with completed birth histories

	2006 sample			Later samples			Test
	N	Mean	SD	N	Mean	SD	
Bottom 2 quintiles	170	0.506	0.501	324	0.454	0.499	F= 1.216
Muslim	170	0.912	0.284	324	0.932	0.252	F= 0.664
Polygamous	170	0.559	0.498	324	0.457	0.499	F= 4.669**
Rank among polygamous wives	95	1.326	0.591	148	1.311	0.558	F= 0.043
Current age - W	170	46.341	1.385	324	46.364	1.348	F= 0.032
Years of education	170	1.024	2.704	324	0.994	2.738	F= 0.013
Received any education	170	0.159	0.367	324	0.151	0.359	F= 0.049
Current age - H	170	53.218	3.962	324	53.213	5.111	F= 0
Age at first birth	166	20.096	4.707	315	21.46	5.845	F= 6.734***
Age at marriage	170	17.135	3.869	324	19.599	5.395	F= 27.906***
Husband's age at marriage	170	23.829	5.009	324	26.12	6.018	F= 18.064***
Has no children	170	0.035	0.185	324	0.037	0.189	F= 0.01
Number of children (ever born)	170	7.388	3.044	324	6.469	3.036	F= 10.2***
Number of children (living)	170	5.247	2.444	324	5.488	2.521	F= 1.037
Ideal number of children - H	132	10.939	6.54	249	10.273	6.458	F= 0.91
Ideal number of children - W	116	7.293	2.601	286	6.706	3.001	F= 3.398*
Ideal number of boys	116	3.448	1.899	285	3.119	1.871	F= 2.527
Ideal numbers of girls	116	3.069	1.758	285	2.67	1.667	F= 4.571**
Ideal number is zero - W	116	0	0	286	0.038	0.193	F= 4.617**

Statistical significance markers: * p<0.1; ** p<0.05; *** p<0.01.

Table 16: Marital characteristics and fertility preferences by DHS wave

Waves Variable	N	2006			2012-2013			2018			Test
		Mean	SD	N	Mean	SD	N	Mean	SD		
Bottom 2 quintiles of wealth index	2665	0.403	0.491	3039	0.418	0.493	2954	0.39	0.488	F= 2.428*	
Muslim	2639	0.914	0.281	3039	0.924	0.265	2954	0.952	0.213	F= 17.635***	
Polygamous	2665	0.385	0.487	3039	0.344	0.475	2954	0.316	0.465	F= 15.079***	
Rank among polygamous wives	1027	1.594	0.632	1046	1.553	0.592	933	1.526	0.558	F= 3.234**	
Current age - wife	2665	29.745	8.283	3039	29.933	7.923	2954	29.634	8.064	F= 1.049	
Years of education	2658	0.845	2.409	3039	1.222	3.063	2954	1.869	3.483	F= 82.251***	
Received any education	2665	0.152	0.359	3039	0.176	0.381	2954	0.269	0.444	F= 70.011***	
Current age - husband	2665	40.745	9.175	3039	40.354	9.55	2954	39.568	9.456	F= 11.544***	
Age at first birth	2483	18.134	3.475	2873	18.847	4.184	2729	18.763	3.749	F= 26.809***	
Age at marriage	2665	16.349	3.246	3039	17.369	4.212	2954	17.374	3.887	F= 65.982***	
Husband's age at marriage	2665	27.294	6.72	3039	27.817	6.85	2954	27.292	6.579	F= 5.994***	
Has no children	2665	0.089	0.285	3039	0.065	0.247	2954	0.084	0.278	F= 6.317***	
Number of children (ever born)	2665	4.321	2.922	3039	3.897	2.502	2954	3.869	2.644	F= 24.615***	
Number of children (living)	2665	3.321	2.242	3039	3.404	2.146	2954	3.377	2.21	F= 1.047	
Ideal number of children - husband	2122	9.242	5.286	2340	9.238	5.676	2543	8.951	5.793	F= 2.148	
Ideal number of children	2103	6.554	2.433	2950	6.148	2.415	2611	6.056	2.601	F= 25.974***	
Ideal number of boys	2100	3.267	1.654	2950	2.963	1.634	2605	2.802	1.717	F= 45.885***	
Ideal numbers of girls	2100	2.819	1.494	2950	2.54	1.467	2605	2.464	1.477	F= 36.393***	
Ideal number is zero - W	2103	0.001	0.038	2950	0.015	0.121	2611	0.034	0.182	F= 37.772***	

Statistical significance markers: * p<0.1; ** p<0.05; *** p<0.01

Table 17: Marital characteristics and fertility preferences by polygamy status - for women aged 45+

Wave	2006			2012-2013			2018			
Variable	N	Mean	SD	N	Mean	SD	N	Mean	SD	Test
Bottom 2 quintiles of wealth index	170	0.506	0.501	176	0.432	0.497	148	0.48	0.501	F= 0.977
Muslim	170	0.912	0.284	176	0.915	0.28	148	0.953	0.213	F= 1.166
Polygamous	170	0.559	0.498	176	0.455	0.499	148	0.459	0.5	F= 2.334*
Rank among polygamous wives	95	1.326	0.591	80	1.375	0.624	68	1.235	0.461	F= 1.127
Current age - wife	170	46.341	1.385	176	46.182	1.305	148	46.581	1.37	F= 3.518**
Years of education	170	1.024	2.704	176	0.915	2.565	148	1.088	2.937	F= 0.168
Received any education	170	0.159	0.367	176	0.136	0.344	148	0.169	0.376	F= 0.35
Current age - husband	170	53.218	3.962	176	53.42	5.144	148	52.966	5.077	F= 0.368
Age at first birth	166	20.096	4.707	174	21.54	5.93	141	21.362	5.759	F= 3.402**
Age at marriage	170	17.135	3.869	176	20.153	5.86	148	18.939	4.718	F= 16.528***
Husband's age at marriage	170	23.829	5.009	176	27.04	6.276	148	25.027	5.52	F= 14.323***
Has no children	170	0.035	0.185	176	0.017	0.13	148	0.061	0.24	F= 2.204
Number of children (ever born)	170	7.388	3.044	176	6.403	2.897	148	6.547	3.201	F= 5.182***
Number of children (living)	170	5.247	2.444	176	5.58	2.451	148	5.378	2.607	F= 0.779
Ideal number of children - husband	132	10.939	6.54	130	10.269	6.566	119	10.277	6.365	F= 0.454
Ideal number of children	116	7.293	2.601	170	6.694	2.997	116	6.724	3.02	F= 1.698
Ideal number of boys	116	3.448	1.899	170	3.088	1.777	115	3.165	2.008	F= 1.318
Ideal numbers of girls	116	3.069	1.758	170	2.712	1.687	115	2.609	1.642	F= 2.408*
Ideal number is zero - W	116	0	0	170	0.035	0.185	116	0.043	0.204	F= 2.383*

Statistical significance markers: * p<0.1; ** p<0.05; *** p<0.01

Table 18: Change in first wives' spacing: duration model

	Hazard ratios			
	Whole sample	Below median reproductive period	Above median	
	(1)	(2)	(3)	(4)
After	0.108** (0.016, 0.199)	0.157* (0.015, 0.298)	0.094 (−0.027, 0.214)	1.350 (−0.364, 3.060)
After × cowife's reproductive period				−0.042 (−0.100, 0.015)
Controls	<i>Birth rank dummies, mother's age and age² at birth, wave FE</i>			
Strata	<i>Husband's and first wife's age in 5 year bands, and woman-level dummy</i>			
Observations	25,681	4,470	2,495	2,495
R ²	0.059	0.067	0.062	0.062

*Cox analysis with proportional hazards, stratified at the woman level, and for categorical ages of the husband and the wife. Sample includes all first and monogamous wives under 45 years old. Confidence intervals are given at the 10% confidence level. Standard errors are robust and clustered at the census enumeration areas. All specifications include birth rank dummies, age and age squared at first birth for the first wife, year and fixed effects for the different survey rounds are included. After refers to a dummy that takes the value 1 at the birth interval during which the second wife enters the household. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 19: Change in first wives' birth spacing: duration model

	Whole sample	Natural/strategic	Poor	Husband's wealth			Wife's educated	
	(1)	(2)	(3)	Middle	Rich	No	Yes	(7)
After	0.108** (0.016, 0.199)	0.408 (-0.384, 1.200)	-0.383 (-1.910, 1.140)	0.509 (-0.555, 1.570)	1.400 (-0.507, 3.300)	0.300 (-0.533, 1.130)	1.420 (-1.300, 4.150)	
After × cowife's RP		-0.010 (-0.038, 0.017)	0.019 (-0.035, 0.072)	-0.014 (-0.051, 0.023)	-0.048 (-0.117, 0.021)	-0.007 (-0.036, 0.022)	-0.042 (-0.138, 0.054)	
Controls	<i>Birth rank dummies, mother's age and age² at birth, wave FE</i>							
Strata	<i>Husband's and first wife's age in 5 year bands, and woman-level dummy</i>							
Observations	25,776	4,580	993	3,220	367	4,166	414	
R ²	0.017	0.036	0.046	0.036	0.098	0.041	0.056	

*Cox analysis with proportional hazards. Sample includes all first and monogamous wives under 45 years old. Standard errors are robust and clustered at the census enumeration areas. Confidence intervals are given at the 10% confidence level. After refers to a dummy that takes the value 1 at the birth interval during which the second wife enters the household. Columns 3 to 7 also include observations which correspond to the header: Husband's wealth is based on the husband's wealth index, which is divided into quintiles: "Poor" for the bottom quintile, "Rich" for the top quintile and "Middle" for the 3 remaining quintiles. Wife's education is indicated by a dummy which takes the value 1 if the first wife has received any education. Signif. Codes: ***, 0.01, **, 0.05, *, 0.1*

Table 20: Comparison of birth intervals including ages

	Birth intervals (in months)			
	Whole sample	Below median	Above median	
		reproductive period		
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Polygamous stage	0.0564 (0.5304)	0.7457 (0.7684)	-1.201 (0.9262)	11.38*** (3.197)
Polygamous stage × cowife's reproductive period				-0.4171*** (0.1119)
Control mean	31.1	30.6	31.2	31.1
<i>Fixed-effects</i>				
Wave variable	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,606	3,591	2,015	3,681
R ²	0.01795	0.02261	0.01755	0.02568
Within R ²	0.00856	0.01007	0.01289	0.01827

*Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Sample for this regression is all senior wives below 45 years old, with at least one child. All specifications include birth rank dummies, age and age squared at first birth for the first wife, categorical variables of the current ages of the husband and the wife. Fixed effects for the different survey rounds are included. Polygamous stage refers to a dummy that takes the value 1 after the entry of the second wife in the household. Signif. Codes: ***, 0.01, **, 0.05, *, 0.1*

Table 21: Comparison of birth intervals

	Birth intervals (in months)			
	Whole sample	Below median reproductive period	Above median	
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Polygamous stage	0.0944 (0.5298)	0.8398 (0.7628)	-1.028 (0.9236)	-0.5630 (0.7376)
Polygamous stage $\times$ Cowife's low ideal				1.857** (0.8135)
<i>Fixed-effects</i>				
Wave variable	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,606	3,591	2,015	3,236
R ²	0.01732	0.02197	0.01616	0.02297
Within R ²	0.00792	0.00942	0.01149	0.01474
<i>Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Sample for this regression is all senior wives below 45 years old, with at least one child. All specifications include birth rank dummies, age and age squared at first birth for the first wife, categorical variables of the current ages of the husband and the wife. Fixed effects for the different survey rounds are included. Polygamous stage refers to a dummy that takes the value 1 after the entry of the second wife in the household. "Low ideal" is a dummy for a cowife who reported her ideal fertility preferences, as below or equal to 5. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>				

Table 22: Predictors of fertility (dep.var: number of children per wife)

	Ever born (1)	Living (2)
<i>Variables</i>		
Senior wife	-0.0328 (0.2975)	-0.0452 (0.2364)
Junior wife	-1.051*** (0.3632)	-1.079*** (0.2986)
Years of education	-0.0555** (0.0217)	-0.0416** (0.0174)
Age at first birth	-0.1348*** (0.0157)	-0.0865*** (0.0138)
Age at marriage - W	0.0880*** (0.0176)	0.0571*** (0.0153)
Age at marriage - H	-0.0645*** (0.0081)	-0.0452*** (0.0070)
Ideal number of children - H	0.0974*** (0.0114)	0.0676*** (0.0101)
Junior wife $\times$ Ideal number of children - W	-0.0217 (0.0512)	-0.0129 (0.0410)
Ideal number of children - W	0.1651*** (0.0265)	0.1169*** (0.0212)
Frequency intercourse past month	0.0137** (0.0068)	0.0144** (0.0059)
Survey wave fixed effects	Yes	Yes
<i>Fit statistics</i>		
Observations	1,841	1,841
R ²	0.26014	0.21811
Within R ²	0.25078	0.21804
<i>Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration area level. "H" and "W" indicate that these outcomes are referring to husbands, and wives respectively. "Ideal number of children - H" is the ideal number of children as expressed by the husband. Fixed effects for the survey waves are included, as well as controls for being Muslim and being part of the two bottom quintiles of the wealth index distribution.</i>		
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Table 23: Fertility patterns for women with complete birth histories

Dependent Variable:	Number of living children			
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Polygamous	-0.6239** (0.2570)			
Senior wife		-0.2943 (0.2826)	-0.2125 (0.2615)	-0.2658 (0.2561)
Junior wife		-0.8333** (0.3892)	-0.8521** (0.3862)	-0.7954** (0.3856)
Reproductive period				0.0783*** (0.0232)
Control mean	5.750	5.750	5.840	5.840
Observations	494	494	476	476
R ²	0.01734	0.01279	0.01530	0.03832
Within R ²	0.01603	0.01147	0.01330	0.03637
<i>Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Fixed effects for the different survey rounds are included. Columns 3 and 4 only include women who have at least one child, discarding infertile unions. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. Survey weights are included.</i>				

Table 24: Fertility for women who completed their birth histories (aged 45 years old and older), by DHS wave

Dependent Variable:	Number of living children					
	2006 sample		2012-2013 sample		2018 sample	
Polygamous	-1.493*** (0.3971)		0.3619 (0.3952)		-0.6982 (0.4782)	
Senior wife		-1.101*** (0.3892)		0.7869* (0.4222)		-0.4106 (0.4602)
Junior wife		-1.556** (0.6734)		0.0026 (0.5449)		-0.8963 (0.7655)
Reproductive period		0.0522 (0.0474)		0.0840** (0.0340)		0.0690* (0.0406)
Constant	6.124*** (0.2909)	4.636*** (1.291)	5.404*** (0.2746)	3.307*** (0.8740)	5.744*** (0.3284)	4.167*** (1.050)
Sampling weights	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	170	164	176	173	148	139
R ²	0.10142	0.08978	0.00563	0.07351	0.01736	0.02774
Adjusted R ²	0.09607	0.07272	-0.00008	0.05706	0.01063	0.00614
<i>Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Fixed effects for the different survey rounds are included. Columns 2, 4 and 6 only include women who have at least one child, discarding infertile unions. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. Survey weights are included.</i>						

Table 25: Comparison of birth intervals including flexible age controls

	Birth intervals (in months)			
	Whole sample	Below median reproductive period	Above median	
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Polygamous stage	0.1422 (0.7510)	1.474 (1.600)	0.0237 (1.443)	13.64*** (4.396)
Polygamous stage $\times$ cowife's reproductive period				-0.4751*** (0.1554)
Control mean	31.1	30.6	31.2	31.1
Wave fixed effect	Yes	Yes	Yes	Yes
Sampling weights	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	6,855	1,892	2,018	4,464
R ²	0.03739	0.06799	0.03740	0.04184
Within R ²	0.03227	0.06277	0.03606	0.03804

*Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration areas. Sample for this regression is all senior wives below 45 years old, with at least one child. All specifications include birth rank dummies, age and age squared at first birth for the first wife, categorical variables of the current ages of the husband and the wife in 5-year bands. Fixed effects for the different survey rounds are included. Polygamous stage refers to a dummy that takes the value 1 after the entry of the second wife in the household. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. Survey weights are included.*

Table 26: Predictors of fertility (dep.var: number of children per wife)

	Ever born (1)	Living (2)
<i>Variables</i>		
Senior wife	-0.1341 (0.2485)	-0.0594 (0.2099)
Junior wife	-1.230*** (0.2330)	-1.028*** (0.1989)
Years of education	-0.0668*** (0.0187)	-0.0560*** (0.0155)
Age at first birth	-0.1533*** (0.0138)	-0.1029*** (0.0118)
Age at marriage - W	0.0829*** (0.0147)	0.0584*** (0.0130)
Age at marriage - H	-0.0715*** (0.0099)	-0.0507*** (0.0078)
Ideal number of children - H	0.0759*** (0.0086)	0.0557*** (0.0069)
Junior wife $\times$ Ideal nb. - W	0.0019 (0.0030)	-0.0008 (0.0024)
Ideal number of children - W	0.1625*** (0.0215)	0.1170*** (0.0178)
Frequency intercourse past month	0.0096 (0.0062)	0.0120** (0.0051)
Survey wave fixed effects	Yes	Yes
Sampling weights	Yes	Yes
<i>Fit statistics</i>		
Observations	3,002	3,002
R ²	0.37510	0.34765
Within R ²	0.37215	0.34399

Heteroskedasticity-robust standard-errors in parentheses. Standard errors are clustered at the census enumeration area level. "H" and "W" indicate that these outcomes are referring to husbands, and wives respectively. "Ideal number of children - H" is the ideal number of children as expressed by the husband. Fixed effects for the survey waves are included, as well as controls for being Muslim and being part of the two bottom quintiles of the wealth index distribution. Missing values for ideal number of children (for both men and women), age at first birth and frequency of intercourse are imputed, and indicator variables for missingness in these variables are included in each regression. Survey weights are included.

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*